I wrote it by hand, but when submitting the assignment on the Moodle website, it said it had to contain at least 20 characters, so I typed this page to make up for those 20 characters.

I \ II	A (q)	B ($1-q$)	
1	b, 0	0, a	p
2	0, 0	d, c	$1-p$

(I) Let player I play 1 chance $= p$

player II play A chance $= q$

For player I $\begin{cases} \text{choose } 1 & U_I(1) = \underline{b}\cdot q + 0\,(\underline{1}-q) = bq \\ \text{choose } 2 & U_I(2) = 0\cdot q + d(1-q) = d - dq \end{cases}$

For player II $\begin{cases} \text{choose } A & U_{II}(A) = 0\cdot p + 0\cdot(1-p) = 0 \\ \text{choose } B & U_{II}(B) = a\cdot p + c(1-p) = ap + c(1-p) \end{cases}$

In Mixed NE

$$\begin{cases} bq = d - dq \\ 0 = ap + c(1-p) \end{cases} \Rightarrow \boxed{\begin{cases} q = \dfrac{d}{b+d} \ (b+d \neq 0), & q \in (0,1), \ bd > 0 \\ p = \dfrac{-c}{a-c} \ (a \neq c), & p \in (0,1), \ ac < 0 \end{cases}}$$

※ (I)

(II)

For player I mix strategy $(\frac{1}{2}, \frac{1}{2})$ means $\Rightarrow p = \frac{1}{2}$, $1-p = \frac{1}{2}$

$\Rightarrow U_{II}(B) = a\cdot\frac{1}{2} + c\cdot\frac{1}{2} = \frac{a+c}{2}$, $U_{II}(A) = 0$

For best response of player II is $(1,0) \Rightarrow$ choose A so $U_{II}(A) > U_{II}(B) \Rightarrow 0 > \frac{a+c}{2}$

$\Rightarrow \boxed{a+c < 0}$ ※ (II)

(III) $\boxed{\text{choose } a=0,\ c=0,\ b=1,\ d=1}$ ※ (III)

For player II is always indifferent

$U_{II}(A) = 0\cdot p + 0(1-p) = 0$ $U_{II}(B) = ap + c(1-p) = 0$.

$U_I(1) = bq = q$ $U_I(2) = d - dp = 1 - q$ $\Rightarrow$ For player I $q = 1-q \Rightarrow q = \frac{1}{2}$ is indifferent

If $q > \frac{1}{2} \Rightarrow q > 1-q \Rightarrow$ choose 1 $\Rightarrow p = 1$

$q < \frac{1}{2} \Rightarrow q < 1-q \Rightarrow$ choose 2 $\Rightarrow p = 0$

$q = \frac{1}{2} \Rightarrow q = 1-q \Rightarrow$ indifferent $\Rightarrow p \in [0,1]$

so for Nash equilibria

$$\begin{cases} (1, q), & q \in [\frac{1}{2}, 1] \\ (0, q), & q \in [0, \frac{1}{2}] \\ (p, \frac{1}{2}), & p \in [0,1] \end{cases}$$

(IV) pure NE fixed player I to 2, player II to B

$\boxed{\text{choose } a=-1,\ c=1,\ b=-1,\ d=2}$ ※ (IV)

$U_{II}(A) = 0$ $U_{II}(B) = 1-2p$

$1-2p \begin{cases} > 0 \Rightarrow & < \frac{1}{2} \rightarrow B \\ = 0 \Rightarrow & p > \frac{1}{2} \rightarrow A \\ < 0 \Rightarrow & = \frac{1}{2} \rightarrow \text{indifferent} \end{cases}$

$U_I(1) = -q$ $U_I(2) = 2-2q$

$\begin{cases} q > 2-2q \Rightarrow q > 2 \ (\times) \\ q < 2-2q \Rightarrow q < 2 \rightarrow 2 \end{cases}$

$\Rightarrow$ player I choose 2 then $p=0$

put back player II

$U_{II}(A) = 0$ $U_{II}(B) = 1$

$\Rightarrow$ pure Nash equilibria

$(2, B)$